DL Class Project Report

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0.1. Introduction

The following is a report about the study of various feature detectors and image matching algorithms applied on the forehead dataset. The project covers the use of SIFT, SURF, ORB, ArcFace, and AKAZE algorithms for matching images. Image Preprocessing (CLAHE) was performed before applying the matching algorithms. A genuine-imposter histogram was plotted for each algorithm. The generated genuine and histogram datasets were used to plot the receiver operating characteristic (ROC) curve of FRR (X-Axis) vs FRR (Y-Axis). Correct Recognition Rate (CRR) and Equal Error Rate (EER) were calculated from the resulting ROC curve.

0.2. Dataset

The dataset consisted of 336 images of 28 subjects, with each subject contributing 12 images over the course of 2 sessions separated by 12+ hours. Some of the images were taken in dim light, whereas some were taken in bright conditions. Thus, data preprocesing was required before applying the feature matching algorithms. Further, all the algorithms gave a very high peak at the matching score of 0. To present the results graphically (and ease of viewing the plots for analytical purposes), the matches with score less than 0.01 and score greater than 2.0 were ignored. This study reports the performance of the algorithms on the raw as well as preprocessed data.

0.3. Data Pre-processing

Data was collected using the front camera of an android phone, executed using an application. The Region of Interest (ROI) was marked on the screen using a rectangular box, so that the user can orient the front camera to fit the forehead in the box. The obtained images were enhanced using contrast limited adaptive histogram equalization (CLAHE) for better visual clarity.

0.4. Key Metric

CRR, EER, and Accuracy are calculated for each matching algorithm. The performance is evaluated based on the CRR and EER. CRR is calculated as follows:

$$CRR(\%) = \frac{(TP + TN)}{N} * 100$$

Here, TP is true positive, and is obtained from the genuine samples of the data. TN is true negative, and is obtained from the imposter samples of the data. N is the total number of samples, and is the sum of total number of samples in genuine and imposter data for that algorithm. FAR and FRR were calculated for each algorithm. The EER was calculated for FAR and FRR values when the absolute difference between the two was minimum (as getting FAR and FRR exactly equal is difficult). EER is given as:

$$EER(\%) = \frac{FAR + FRR}{2}$$

Accuracy is given by:

$$Accuracy(\%) = 100 - \frac{(FAR + FRR)}{2}$$

So, in this case, accuracy can also be given as:

$$Accuracy(\%) = 100 - EER$$

0.5. Algorithms

The use of SIFT, SURF, ORB, ArcFace, and AKAZE algorithms for matching images is studied. Image Preprocessing (CLAHE) was performed before applying the matching algorithms. The Genuine - Imposter histogram has been plotted for all the algorithms, with probability*100 plotted on the Y-Axis.

0.5.1 SIFT:

Scale-Invariant Feature Transform, or, SIFT transforms image data into linear scale-invariant coordinates that are relative to local features. These features are invariant to image translation, scaling and rotation and robust to change in illumination and viewpoint. The algorithm generates a large number of features that cover the image[1] using blob detection. For object matches to be verified there has to be at least 3 features that perfectly agree on an object.

Without Image Enhancing

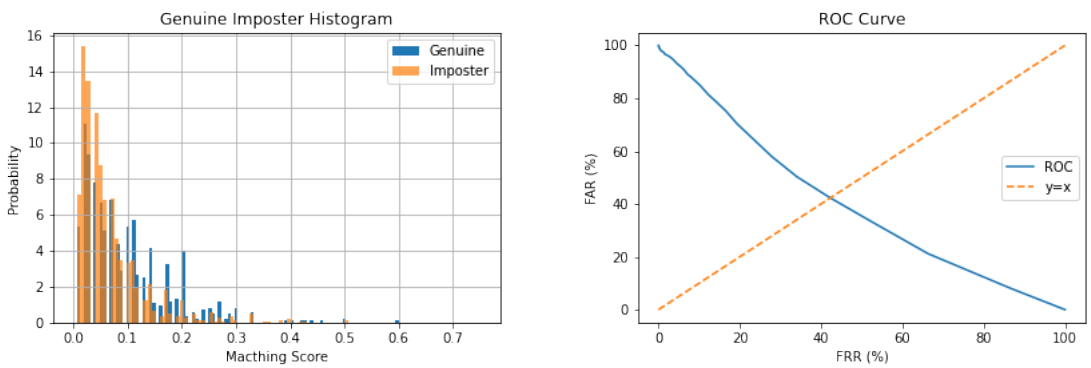

Figure 1: SIFT: without CLAHE

The length of the genuine data is 819, and that of the imposter matching data is 11428. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 56.634%, and the EER was 41.996%, both for the threshold of 0.050. The Accuracy was 58.004%.

With Image Enhancing

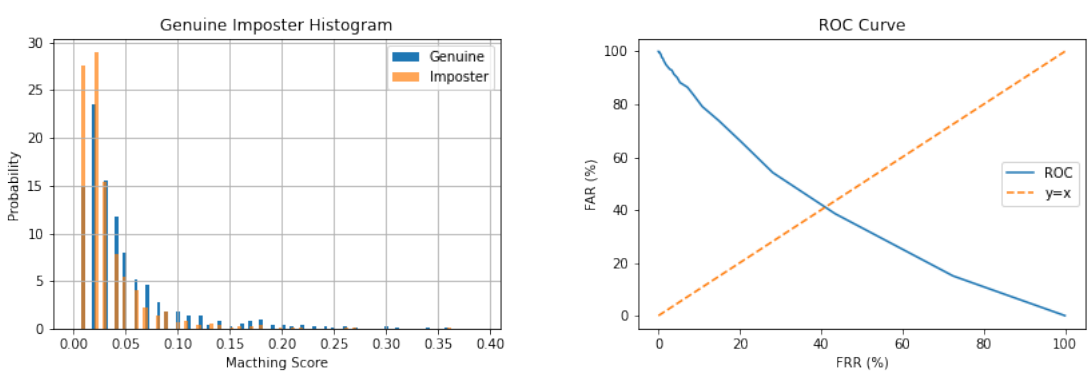

Figure 2: SIFT: with CLAHE

The length of the genuine data is 1164, and that of the imposter matching data is 17831. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 56.825%, and the EER was 41.024%, both for the threshold of 0.020. The Accuracy was 58.976%.

Ablation Study: Changing Hyper-Parameters:

In the ratio test for matching between the two images, Lowe's ratio was originally 0.7 (for the results shown above). Experiments were performed by changing this value, and results were obtained for Lowe's ratio = 0.8, the CRR

obtained was 58.965%, and the EER was 42.185%, both for the threshold of 0.090. The Accuracy was 57.815%. For Lowe's ratio = 0.9, the CRR obtained was 55.438%, and the EER was 45.071%, both for the threshold of 0.240. The Accuracy was 54.919%. These results were obtained on raw data (unprocessed images).

0.5.2 SURF:

Speeded Up Robust Features, or SURF, is a fast and robust algorithm for local, similarity invariant representation and comparison of images. SURF detector focuses its attention on blob-like structures in the image. These structures can be found at corners of objects, but also at locations where the reflection of light on specular surfaces is maximal (i.e. light speckles)

Without Image Enhancing

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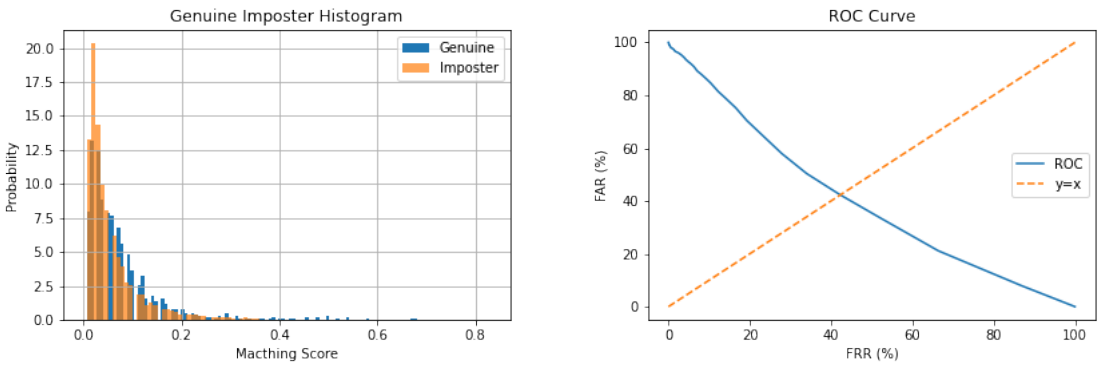

Figure 3: SURF: Without CLAHE

The length of the genuine data is 1561, and that of the imposter matching data is 32753. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 57.898%, and the EER was 42.279%, both for the threshold of 0.040. The Accuracy was 57.721%.

With Image Enhancing

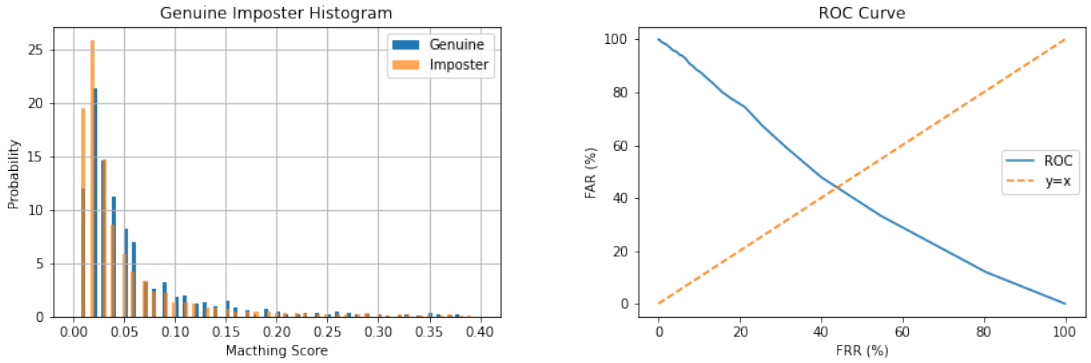

Figure 4: SURF: With CLAHE

The length of the genuine data is 1522, and that of the imposter matching data is 32116. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 59.626%, and the EER was 43.957%, both for the threshold of 0.030. The Accuracy was 56.043%.

Ablation Study: Changing Hyper-Parameters:

In the ratio test for matching between the two images, Lowe's ratio was originally 0.7 (for the results shown above). Experiments were performed by changing this value, and results were obtained for Lowe's ratio 0.8 and 0.9. For Lowe's ratio = 0.8, the CRR obtained was 58.450%, and the EER was 42.269%, both for the threshold of 0.100. The Accuracy was 57.731%. For Lowe's ratio = 0.9, the CRR obtained was 56.380%, and the EER was 44.397%, both for the threshold of 0.300. The Accuracy was 55.603%. These results were obtained on raw data (unprocessed images).

0.5.3 ORB:

ORB (oriented BRIEF) is another key-point detector and descriptor extractor available in the OpenCV library. The Oriented Fast and Rotated Brief (ORB) algorithm is based on the BRIEF key-point descriptor and the FAST key-point detector since both algorithms are computationally fast. The following results are obtained keeping the default ORB parameters intact. Since ORB is a binary descriptor, the Hamming Norm was used for the Brute Force Matcher. The default number of features are 500.

Without Image Enhancing

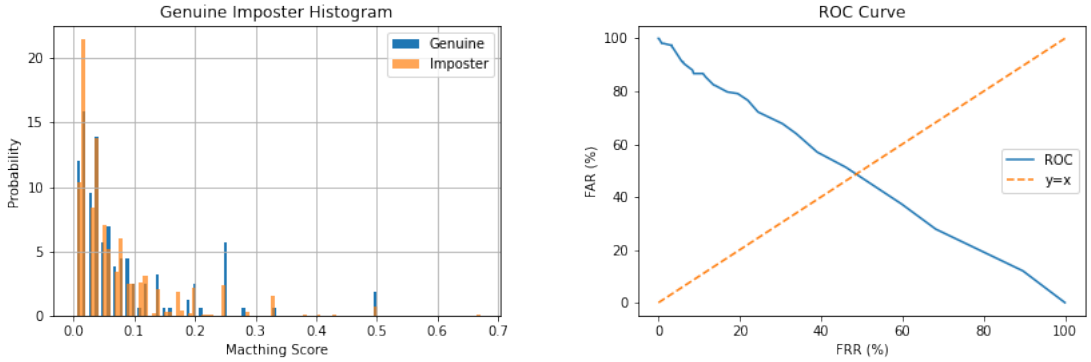

Figure 5: ORB: Without CLAHE

The length of the genuine data is 158, and that of the imposter matching data is 1943. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 53.546%, and the EER was 48.664%, both for the threshold of 0.190. The Accuracy was 51.336%.

With Image Enhancing

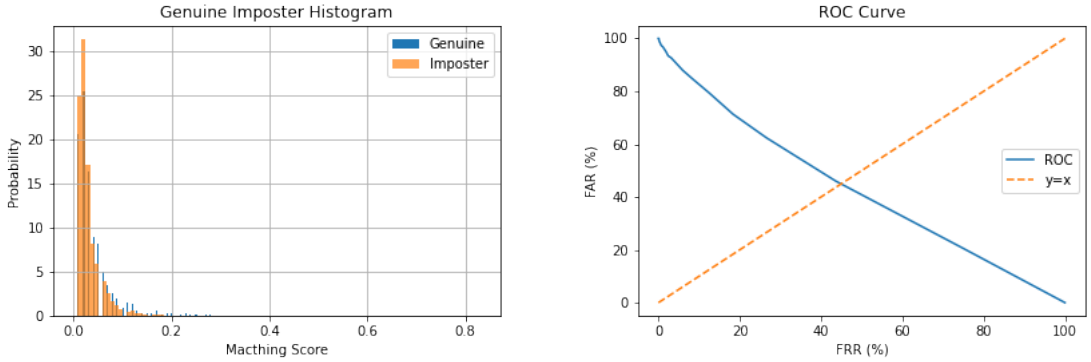

Figure 6: ORB: With CLAHE

The length of the genuine data is 837, and that of the imposter matching data is 16398. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 56.252%, and the EER was 44.815%, both for the threshold of 0.020. The Accuracy was 55.185%.

Ablation Study: Changing Number of Features:

When the number of features are increased to 10000, the following results are obtained:

Without Image Enhancing

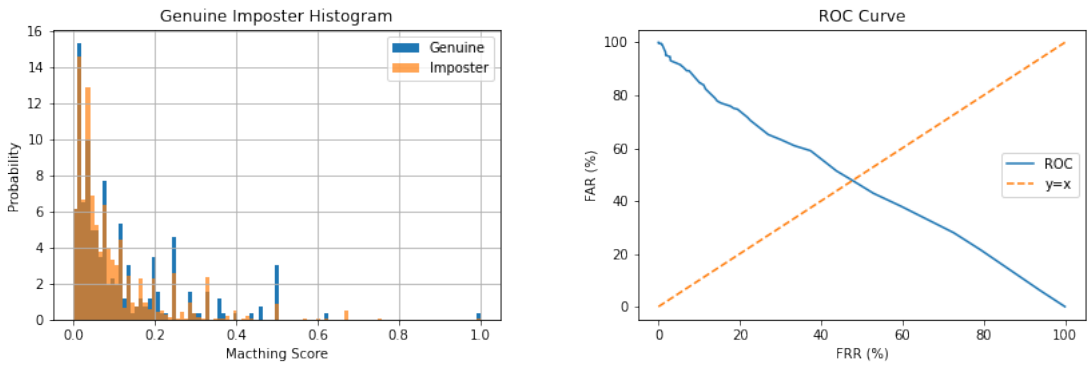

Figure 7: ORB: Nfeatures = 10,000 (Without CLAHE)

The length of the genuine data is 261, and that of the imposter matching data is 3713. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 52.391%, and the EER was 47.741%, both for the threshold of 0.761. The Accuracy was 52.259%.

With Image Enhancing

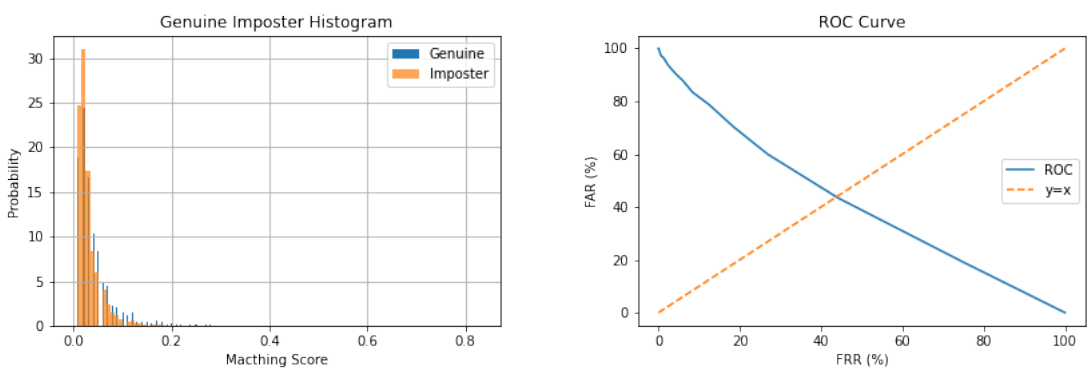

Figure 8: ORB: Nfeatures = 10,000 (With CLAHE)

The length of the genuine data is 761, and that of the imposter matching data is 15634. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 55.810%, and the EER was 43.797%, both for the threshold of 0.020. The Accuracy was 56.203%.

0.5.4 ArcFace

ArcFace face recognition algorithm was used to obtain the matching scores between two images. ArcFace face recognition implementation in Tensorflow Lite (available as a pypi package) was used for this project. It included a pre-trained model based on ResNet50. The matching scores are inversely proportional to how good the matching is between the two images. Hence, the method of finding FAR and FRR is opposite to the others.

Without Image Enhancing

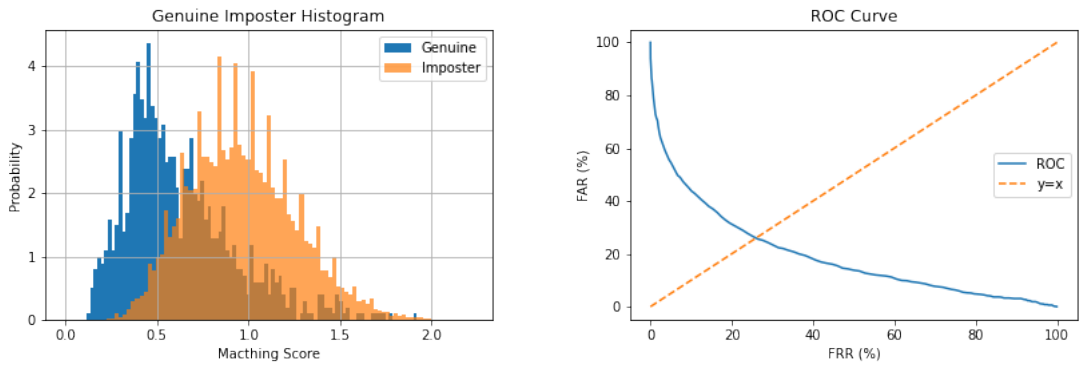

Figure 9: ArcFace: Without CLAHE

The length of the genuine data is 2016, and that of the imposter matching data is 54432. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 26.382%, and the EER was 26.003%, both for the threshold of 0.761. The Accuracy was 73.997%.

With Image Enhancing

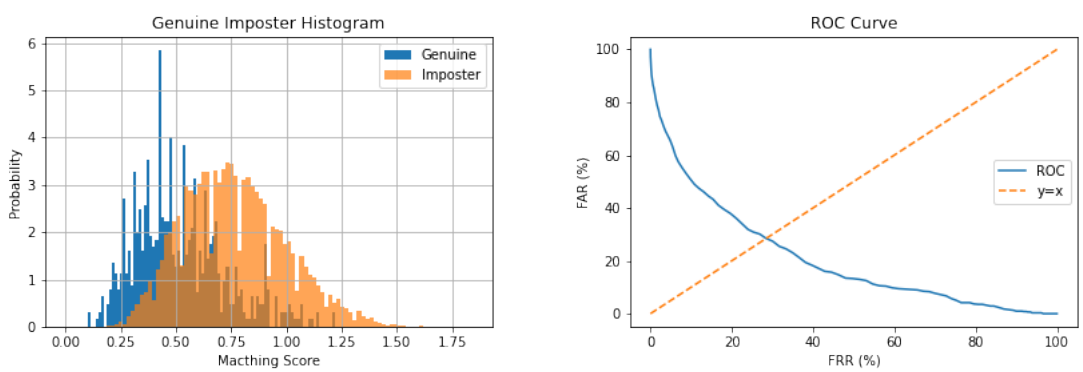

Figure 10: ArcFace: With CLAHE

The length of the genuine data is 1248, and that of the imposter matching data is 33712. Some matches were ignored as the descriptor returned by the matching algorithm was *None*. Further, some matches returned did not have enough values, and these cases were also ignored. The CRR obtained was 28.410%, and the EER was 28.465%, both for the threshold of 0.611. The Accuracy was 71.535%.

0.5.5 AKAZE:

Accelerated KAZE, or AKAZE, uses a binary descriptor to further increase speed as compared to the KAZE algorithm. The idea behind the creation of this algorithm was to detect and describe 2D features in a nonlinear scale-space extrema so as to obtain a better localization accuracy and distinctiveness.

Without Image Enhancing

AKAZE performed very poorly on unprocessed images. It could obtain just 2 genuine matching scores, and just 24 imposter matching scores. The CRR obtained was 11.538%, and the EER was 93.750%, both for the threshold of 0.250. The Accuracy was 6.250%.

With Image Enhancing

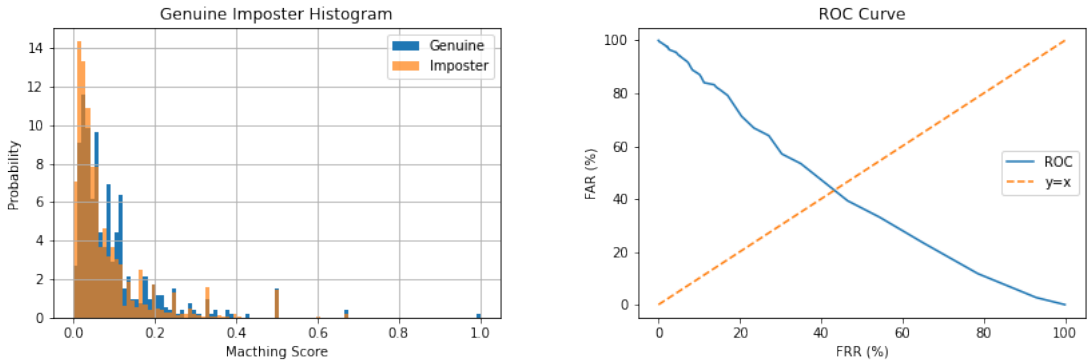

Figure 11: AKAZE: With CLAHE

After applying CLAHE on the images, the performance of AKAZE improved tremendously. It could perform matchings resulting in genuine data with length 519, and imposted data with length 7214. The CRR obtained was 53.938%, EER obtained was 42.927%, both for a threshold of 0.050. The Accuracy was 57.073%.

0.6. Tabulated Results

-	Algorithm	CRR	EER	Accuracy
1	SIFT (without image preprocessing)	56.634%	41.996%	58.004%
2	SIFT(with image preprocessing)	56.825%	41.024%	58.976%
3	SURF (without image preprocessing)	57.898%	42.279%	57.721%
4	SURF(with image preprocessing)	59.626%	43.957%	56.043%
5	ORB (without image preprocessing)	53.546%	48.664%	51.336%
6	ORB(with image preprocessing)	56.252%	44.815%	55.185%
7	ORB with nfeatures = 10000 (without image preprocessing)	52.391%	47.741%	52.259%
8	ORB with nfeatures = 10000 (with image preprocessing)	55.810%	43.797%	56.203%
9	ArcFace (without image preprocessing)	26.382%	26.003%	73.997%
10	ArcFace(with image preprocessing)	28.410%	28.465%	71.535%
11	AKAZE (without image preprocessing)	11.538%	93.750%	6.250%
12	AKAZE (with image preprocessing)	53.938%	42.927%	57.073%

0.7. Conclusion

This report discussed the use of forehead images for human authentication. The data was collected using front camera of a mobile phone, using a custom android application. This made the ROI formation easy for the person clicking images. Various Image Matching and feature extracting algorithms were studied. They were tested on an unprocessed dataset. They were also tested on preprocessed data, where CLAHE was used for histogram equalization of gray scale images. Rotation and scale invariant algorithms like SIFT performed the best among the study, giving an EER of 41.024% on the preprocessed images. It was observed that image enhancing gives better results in all the cases, with the exception of SURF, where the accuracy decreased by about 1.7%. But in that case too, the CRR increased for the image enhanced case by almost 2%. So overall, the effect of CLAHE preprocessing was clearly seen.